

CELEBAL INTERNSHIP ASSIGNMENT – 4

Name – Anmol Bhatia

Dataset - Superstore

Objective

The objective of this assignment is to understand Azure cloud concepts and implement an end-to-end data pipeline using Azure Storage Account and Azure Data Factory. The pipeline validates the source file using the Get Metadata activity and copies the data from the input Blob container to the output Blob container while monitoring the execution.

Resource Group

The screenshot shows the Azure portal interface for a resource group named 'CELEBAL-RG'. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Resource visualizer, Events, Settings, Deployments, Security, Deployment stacks, Policies, Properties, Locks, Cost Management, Cost analysis, Cost alerts (preview), Budgets, Advisor recommendations, Monitoring, Insights (preview), Alerts, Metrics, Diagnostic settings, Logs, and Advisor recommendations. The main content area is titled 'Overview' and includes a search bar and a list of actions: Create, Manage view, Delete resource group, Refresh, Export to CSV, Open query, Assign tags, Move, Delete, Export template, and more. Below this, the 'Essentials' section displays key information: Subscription (linked to 'Azure for Students'), Subscription ID (1e63ffdd-fd1a-424d-bafe-2a9756bd330e), Deployments (No deployments), and Location (East Asia). The 'Tags' section shows 'Add tags'. The 'Resources' section has a filter bar with a search input and filters for 'Type equals all' and 'Location equals all'. A message states 'No resources match your filters' with a 'Create' button and a 'Clear filters' button. A 'Learn more' link is also present. At the bottom, it shows 'Showing 1 - 0 of 0. Display count: auto'.

Storage Account

Home

anmolcestorage

Storage account

Search

Upload

Open in Explorer

Delete

Move

Refresh

Open in mobile

CLI / PS

Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security + networking

Data management

Settings

Monitoring

Automation

Help

Essentials

Resource group (move) : CELEBAL-RG

Location : eastasia

Subscription ID (move) : 6a9e4f4d-fd1a-424d-bafe-2a9756bd330e

Subscription ID : 1e63ffdd-fd1a-424d-bafe-2a9756bd330e

Disk state : Available

Tags (edit) : Add tags

Performance : Standard

Replication : Locally redundant storage (LRS)

Account kind : StorageV2 (general purpose v2)

Provisioning state : Succeeded

Created : 7/11/2026, 11:21:18 AM

Properties

Monitoring

Capabilities (7)

Recommendations (0)

Tutorials

Tools + SDKs

Blob service

Hierarchical namespace : Disabled

Default access tier : Hot

Blob anonymous access : Disabled

Blob soft delete : Enabled (7 days)

Container soft delete : Enabled (7 days)

Versioning : Disabled

Change feed : Disabled

NFS v3 : Disabled

Allow cross-tenant replication : Disabled

Storage tasks assignments : None

File service

Identity-based access : Not configured

Default share-level permissions : Disabled

Soft delete : Enabled (7 days)

Require Encryption in Transit for SMB (Security) : Enabled

Security

Require secure transfer for REST API operations : Enabled

Storage account key access : Enabled

Minimum TLS version : Version 1.2

Infrastructure encryption : Disabled

Networking

Public network access : Enabled

Public network access scope : Enable from all networks

Private endpoint connections : 0

Network routing : Microsoft network routing

Endpoint type : Standard

Blob Containers

Home > anmolcestorage

anmolcestorage | Containers

Storage account

Search

Add container

Upload

Refresh

Delete

Change access level

Restore containers

Edit columns

Search containers by prefix

Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	\$logs	7/11/2026, 11:21:45 AM	Private	Available	-
☐	input	7/11/2026, 1:08:13 PM	Private	Available	-
☐	output	7/11/2026, 1:08:21 PM	Private	Available	-

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security + networking

Data management

Settings

Monitoring

Automation

Help

Source File Uploaded

Home > anmolcestorage | Containers

input

Container

Search

+ Add Directory

Upload

Change access level

Refresh

Delete

Copy

Paste

Rename

Acquire lease

Break lease

Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

input

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	Sample - Superstore.csv	7/11/2026, 1:13:38 PM	Hot (Inferred)	Block blob	2.18 MiB	Available	...

Azure Data Factory Overview

Home

Tools

Monitoring

Settings

Help

Data factory

adf-celebal-anmol

New

Ingest

Copy data at scale once or on a schedule.

Orchestrate

Code-free data pipelines.

Transform data

Transform your data using data flows.

Configure SSIS

Manage & run your SSIS packages in the cloud.

Recent resources

No items to show

Your recently opened resources will show up here.

Discover more

Browse partners (preview)

Pipeline templates

SAP pipeline templates

Linked Service

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

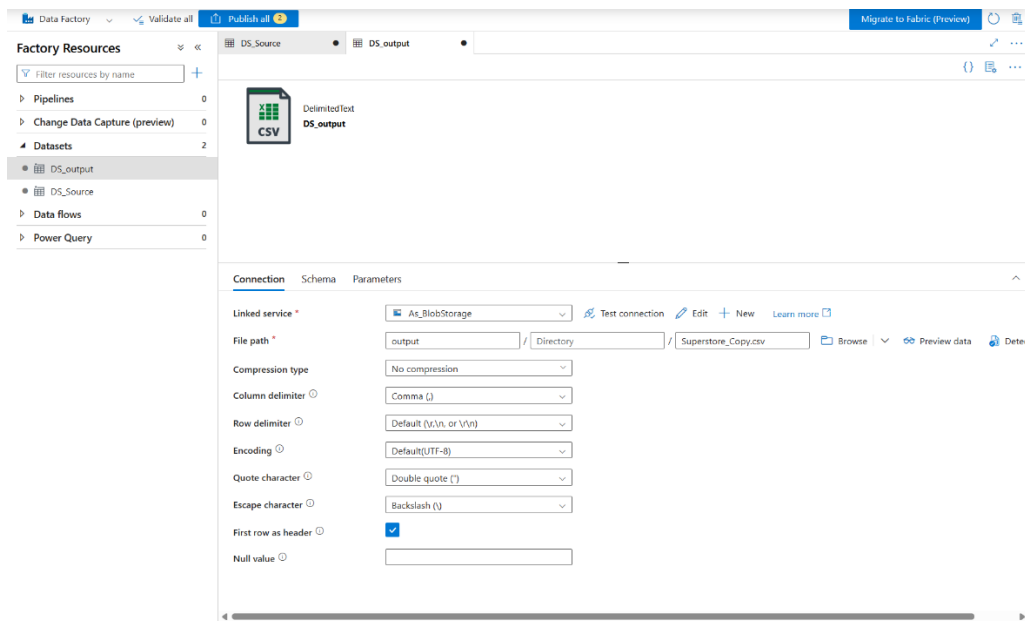
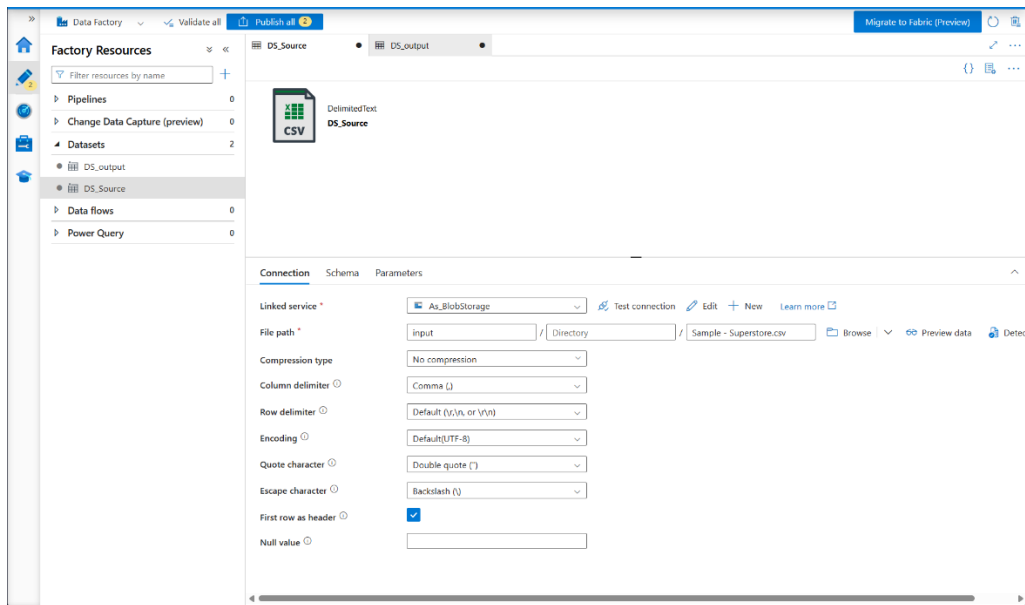
Filter by name

Annotations : Any

Showing 1 - 1 of 1 items

Name	Type	Related	Annotations
As_BlobStorage	Azure Blob Storage	2	

Datasets



Pipeline Design

The screenshot shows the 'Pipeline Design' view in Azure Data Factory. The top navigation bar includes tabs for 'DS_Source', 'DS_output', and 'PL_superstore'. Below the navigation bar, there are buttons for 'Validate', 'Debug', and 'Add trigger'. The left sidebar, titled 'Activities', contains a search bar and a list of activity categories: 'Move and transform' (with 'Copy data' and 'Data flow' sub-items), 'Synapse', 'Azure Data Explorer', 'Azure Function', 'Batch Service', 'Databricks', 'General', 'HDInsight', 'Iteration & conditionals', 'Machine Learning', and 'Power Query'. The main canvas displays a pipeline diagram with two activities: 'Get Metadata' (labeled 'Get Metadata') and 'Copy data' (labeled 'Copy data1'). A green arrow connects the output of 'Get Metadata' to the input of 'Copy data1'. Both activities have a green checkmark in the top right corner, indicating they are valid.

Pipeline Validation

The screenshot shows the 'Pipeline Validation' view in Azure Data Factory. The top navigation bar includes tabs for 'Data Factory', 'Validate all', 'Publish all', and 'Migrate to Fabric (Preview)'. Below the navigation bar, there are buttons for 'Validate', 'Validate copy runtime', 'Debug', and 'Add trigger'. The left sidebar, titled 'Expand toolbox pane', contains a list of activity categories: 'Sink dataset', 'Copy behavior', 'Max concurrent connections', 'Block size (MB)', 'Metadata', 'Quote all text', 'File extension', and 'Max rows per file'. The main canvas displays a pipeline diagram with two activities: 'Get Metadata' (labeled 'Get Metadata') and 'Copy data' (labeled 'Copy data1'). A green arrow connects the output of 'Get Metadata' to the input of 'Copy data1'. The 'Copy data' activity is highlighted with a blue border. The right sidebar, titled 'Pipeline validation output', displays a message: 'Your pipeline has been validated. No errors were found.' Below the message is a 'Close' button.

Debug Execution

Microsoft AzureData Factoryadf-celebal-animol

Search

1000020641@dit.edu.inDIT UNIVERSITY

»Data Factory»Validate allPublish all

DS_SourceDS_outputPL_superstore

ValidateDebugAdd trigger

Get MetadataCopy data

Get MetadataCopy data1

ParametersVariablesSettingsOutput

Pipeline run ID 22a2238d-6575-4188-bae7-1bc1bf448327Pipeline status SucceededView debug run consumption

All status▼Monitor in Azure MetricsExport to CSV

Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
Copy data1	Succeeded	Copy data	7/11/2026, 2:49:37 PM	18s	AutoResolveIntegrationRuntime (East Asia)		bdf59a84-213f-4cb0-ac0a-3c0fcd23777f
Get Metadata	Succeeded	Get Metadata	7/11/2026, 2:49:25 PM	11s	AutoResolveIntegrationRuntime (East Asia)		38cec844-0a0d-473e-a078-415342b4a2

Trigger Execution

»DashboardsRunsPipeline runsTrigger runsChange Data Capture (previ...Runtimes & sessionsIntegration runtimesData flow debugNotificationsAlerts & metrics

Pipeline runsTriggeredDebugRerunCancel optionsRefreshEdit columnsListGantt

Filter by run ID or nameChennai, Kolkata, Mu...Last 24 hoursPipeline name: AllStatus: AllRuns: Latest runsCopy filtersExport to CSV

Triggered by: AllAdd filter

Showing 1 - 1 itemsLast refreshed 0 minutes ago

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameter
PL_superstore	7/11/2026, 3:03:10 PM	7/11/2026, 3:03:56 PM	47s	Manual trigger	Succeeded	Original	

Output Container

Home > anmolcestorage | Containers

output

Container

Search

◁

+ Add Directory

📁 Upload

🔗 Change access level

🔄 Refresh

🗑️ Delete

📄 Copy

📄 Paste

🏷️ Rename

🔑 Acquire lease

🔑 Break lease

🔧 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

output

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	Superstore_Copy.csv	7/11/2026, 2:49:50 PM	Hot (Inferred)	Block blob	2.18 MiB	Available	...

IAM Role Assignment

Home > anmolcestorage

anmolcestorage | Access Control (IAM)

Storage account

Search

◁

+ Add

📄 Download role assignments

🔧 Edit columns

🔄 Refresh

🗑️ Delete

🗨️ Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security + networking

Data management

Settings

Monitoring

Automation

Help

Check access

Role assignments

Roles

Deny assignments

Number of role assignments for this subscription

3

4000

Search by name or email

Type: All

Role: All

Scope: All scopes

Group by: Role

All (3)

Job function roles (2)

Privileged administrator roles (1)

☐	Name	Type	Role	Scope	Condition
▼	Owner (1)				
☐	ANMOL BHATIA 1000020641@dit.edu.in	User	Owner	Subscription (inherited)	None
▼	Storage Blob Data Contributor (1)				
☐	adf-celebal-anmol fb06f776-2a35-49b5-bb64-3614...	Managed identity	Storage Blob Data Contributor	This resource	Add
▼	Storage Blob Data Reader (1)				
☐	adf-celebal-anmol fb06f776-2a35-49b5-bb64-3614...	Managed identity	Storage Blob Data Reader	This resource	Add

Showing 1 - 3 of 3 results.

BRIEF SUMMARY

This assignment successfully demonstrated the implementation of an end-to-end Azure Data Factory pipeline. A Resource Group, Storage Account, Blob Containers, Linked Service, and Datasets were created. The Get Metadata activity validated the source file, and the Copy Data activity transferred the CSV file from the input container to the output container. The pipeline was executed successfully using both Debug and Manual Trigger, and its execution was monitored through Azure Data Factory. Appropriate IAM roles were assigned to ensure secure communication between Azure Data Factory and Azure Blob Storage.